VU AI EX

Bt102 objective notes for mid term

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VU Medical Zone (biotechnologists)

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Bt102 objective notes for mid term

From topics 1 to 12

1) Microbes are also called_____?
   a) germs
   b) viruses
   c) microorganisms
   d) both a and c

2) Studying of microbes is called____?
   a) Mycology
   b) Microbiology
   c) Immunology
   d) Epidemiology

3) Which of the following is used to be the main feature prior to the advent of molecular biology.
   a) Morphology
   b) Bacteriology
   c) Mycology
   d) Microbiology

4) _____are two different groups
   a) Bacteria and archea
   b) Bacteria and fungi
   c) Archea and fungi
   d) Viruses and bacteria

5) What are the basis of classification?
   Similarities among them, Morphology used to be the main feature prior to the advent of molecular biology, Biochemical similarities, Genetic similarities & Evolutionary relationships

6) Which one is an example of protists in eucarya?
   a) Slim molds
   b) Protozoa
   c) Algae
   d) All of these

7) Why do we Classify Organisms?

Studying them becomes convenient, Easy Identification, Easy to establish relationship between organisms & Evolutionary relationships can be established

8) What are **Three Domain Classfication (1977)**

a) Bacteria, Archea, Eucarya
b) Slim molds, Protozoa, Algae
c) Eucarya, Slim molds, Protozoa
d) Archea, Eucarya, Slim molds

9) Unicellular form of fingi is called_____ and Multicellular form of fungi is known as _____?
a) Yeast, mold
b) Yeast, mushroom
c) Mold, mushroom
d) Yeast, mold or mushroom

10) _____composed only of circular ssRNA.
a) Bacteria
b) Viriods
c) Viruses
d) Virons

11) _____ need helper viruses for replication and encapsidationand Also called sattelite viruses
a) Bacteria
b) Viriods
c) Virusoids
d) Virons

12) Which one of the following is an infectious proteins
a) Prions
b) Viroids
c) Bacteria
d) Virusoids

13) In naming organisms Linnaeus established system of specific nomenclature in_____?
a) 1786
b) 1996
c) 1536
d) 1735

14) What two names use for Each organism in **Binomial Nomenclature?**
The genus and specific epithet.

15) **Scientific Names Are italicized or underlined. The genus is ____and the specific epithet is_____?**

a) Lowercase, Capitalized

b) Capitalized, lowercase

16) **Escherichia coli** Honors the discoverer, _____it Describes the bacterium's habitat—the large intestine, or colon.

a) Robert hook

b) Theodor secwaan

c) Theodor Escherich

d) franstoro

17) **Robert Hooke** First pulished drawing of microbes refered in____?

a) 1665

b) 1675

c) 1765

d) 1857

18) **Antony van Leeuwenhoek was** First to publish extensive & accurate observations of microbes He called them _____? "animalcules"

19) observations of microbes renewed the controversyreferes to _____

a) Leeuwenhoek

b) John Needham

c) Louis Pasteur

d) All of these

20) **John Needham** Boiled extracts of hay or meat can give rise to microorganisms He boiled mutton broth and then tightly stoppered the bottles in _____?

a) 1875

b) 1748

c) 1859

d) 1487

21) **_____tells** Microbes are present in the air. They can contaminate sterile solutionsin ____?

a) **Louis Pasteur, 1861**

b) **John Needham , 1748**
c) **Robert Koch, 1878**
d) **Edward janner, 1796**

22) In the presence of _____ bacteria grow and change alcohol to acitic acid
a) H2
b) N2
c) O2
d) Co2

23) What discoveries were made in the golden age of microbiology?
A lots of discoveries in microbiology with Pasteur's work In about 60 years, Microbiology became a real science. Spearheaded by Pasteur and Koch Disease agents identified Role of Immunity established Chemical activities of microbes studied Improved culturing techniques introduced Vaccines Surgical techniques developed

24) In 1860 joseph Lister applied _____as treating surgical wounds and hands sanitizing
a) Phenol
b) Benzene

25) Five kingdom classification was proposed by Robbert Whiltaker in _____
a) 1799
b) 1856
c) 1969
d) 1987

26) Alexander flamming discover pencillium from fungus Pencillium chrysogenum in____?
a) 1987
b) 1928
c) 1996
d) 1978

27) Cell wall of Bacteria is made up of____?
a) Peptidoglycan
b) Glycans
c) Peptidogines
d) cellulose

28) _____are also known as mild dews and white rusts.
a) Slime molds
b) Water molds

29) System of scientific nomenclature established by____?
a) Robert hook
b) Ernest hackel
c) Carl linneaous
d) Loice pasture

30) In 1861 The issue of spontaneous generation was resolved by french scientist_____ named as?
a) Louis pasteurs
b) Jhon needham
c) Robert hook
d) Ernest hackel

31) In_____, Pasteur found that silkworm disease caused by a protozoan
a) 1865
b) 1896
c) 1768
d) 1765

32) The use of chemical for treatment is called as____?
chemo theropy

33) Study of fungi is called _____?
Mycology

34) **What is Bacteriology?**
Study of bacteria. Its Deals with:
– Isolation
– Identification
– Characterization
– Classification

35) what is resolution?
Resolution is the ability of the lenses to distinguish between two closely lying objects as separate. See the accompanying diagram for visual concept of resolution. o Light microscope resolving power is 0.2 um.

36) Mention here the Parts of Compound Light Microscope?
Illuminator: a light source
Condenser: Directs the light through the specimen
Objective Lenses: Close to the specimen
Ocular Lens (Eyepiece): Close to the eye

37) What is total magnification?
Magnification of objective lens x magnification of ocular lens

38) _____are mostly low Objective lenses?
   a. 10 X
   b. 20 X
   c. 30 X
   d. 40X

39) _____are mostly high Objective lenses?
   a. 10 X
   b. 20 X
   c. 30 X
   d. 40X

40) Low magnification power is_____ and Oil immersion power is____?
   a. 100,1000
   b. 1000, 100
   c. 10, 100
   d. 100, 10

41) Ability of a medium to bend the light is called ?
   a. Magnification
   b. Refractive Index
   c. Immersion
   d. Condenser

42) In refractive index Light rays move in a straight line through a single medium. To ____contrast, we stain the specimen?
   a. Increase
   b. Decrease
   c. Stop
   d. Make fast

43) Light microscope resolving power is ____ µm. n = refractive index, 1.5 for immersion oil.
   a. 0.2
   b. 1.0
   c. 0.5
   d. 2.0

44) Ordinary light microscope, Stained images are used, and Can resolve _____objects?
   a. 100 nm

b. 200 nm
c. 300 nm
d. 400 nm

45) What are the Types of Microscopes?

Darkfield microscopy, Phase contrast microscopy, Fluorescent microscopy, Transmission electron microscopy, Scanning electron microscopy

46) Darkfield Microscopy used for _____?

a. darkfield condenser having an opaque disk
b. unstained specimens suspended in liquids
c. Light rays in phase or out of phase
d. Both a and b options are correct

47) In _____Light rays can be in phase or out of phase?

a. Darkfield microscopy
b. Phase Contrast Microscopy
c. Fluorescent microscopy
d. Scanning electron microscopy

48) A host of microscopescan be used to observe the microbes for their morphology _____staining.

a. with
b. without
c. Both
d. None of these

49) What is staining?

It is just coloring with a dye.

50) What are the Advantages of Staining?

Increases visibility of microbes, Shape, size and arrangements, Purity or contamination of a culture, Differentiation and classification of microbes, Various parts of bacteria can be detected with staining

51) How to make a smear?

Using a platinum loop, acolony or a drop of broth culture can be smeared into a thin film on a glass slide for making a smear.

Fixing the smear to the slide: The specimen is spread into a thin film (smear).Smear is air-dried. Smear is fixed (attached) to the slide before

staining.Heating the slide is one way of fixing the smear.Methyl alcohol can be used.

Fixing a smear:

Physically attach the specimen onto the slideTo preserve internal and external structures in place

Physical method: Heat

Chemical Methods: Ethanol or formaldehydeStain is applied and then washed off with water. The slide is air dried or blotted. Staining a smear:

52) Salts composed of a -ive& a +ive ion is called____?

a. Smear
b. Stain
c. Chromophore
d. Resolution

53) What is difference b/w basic and acidic dye?

Basic dye: The color is in the positive ion.

Acidic dye: The color is in the negative ion.

54) Bacteria are negatively charged at pH ____?

a. 2
b. 8
c. 6
d. 7

55) Which of the following is example of Basic DyesPositive Staining?

a. Crystal violet, safranin
b. Methylene blue, safranin
c. Malachite green, Safranin
d. All of these options

56) Which of the following is an example of Acidic Dyes Negative Staining?

a. Eosin
b. Acid fuchsin
c. Nigrosin
d. All of these

57) In Simple staining _____stain/stains is used. Example Methylene Blue

a. One

b. Four
c. Six
d. Ten

58) Differential stains reactdifferently with different kinds of bacteria. Most common is Gram Staining developed by ____?

a. Chief bogus
b. Hans Christian
c. Robert Downey
d. H. Shelley

59) Crystal violet and iodine make a complex in the _____?

a. Cytoplasm
b. Nucleus
c. Chromosome
d. Mitochondria

60) Prokaryotes and eukaryotes are chemically ____?

a. Different
b. Similar

From topics 13 to 16

61) Prokaryotes are Unicellular organisms like Bacteria & Archea Although, look similar, their chemical composition is different. Majority of prokaryotes are _____?

a) Bacteria
b) Archea
c) Bacteria& Archea
d) viruses

62) Basic Shape of bacteria is _____?

a) Cocci
b) Bacilli
c) Spirals
d) All of these

63) What are Basic Spiral Shapes of bacteria?

Vibrio: curved rods
Spirillum: Helical but rigid
Spirochete: Helical but flexible Move by axial filaments

64) *Spirillum minus* causes rat bite fever _____wide and ____long.

a) 0.2 µm, 3 -5µm
b) 3 -5µm, –0.2 µm
c) 0.4 µm, 3 -6µm
d) 1.2 µm, 4 -5µm

65) Spiral Spirochete is Helical but flexible it Move by axial filaments their Size is_____ µm long_____ µm wide?
a) 250, 0.1–0.7
b) 5, 0.1–0.9
c) 5 & 250, 0.1–0.6
d) 6.1, 0.6

66) What are the Arrangements of Bacill?
Bacillu, Diplobacilli & Streptobacilli

67) Which one Divided in 3 planes of Bacilli
a) Sarcinae
b) Streptobacilli
c) Diplobacilli
d) All of these

68) Which one Divided in multiple planes
a) Sarcinae
b) Streptobacilli
c) Diplobacilli
d) All of these

69) Explain the Shapes of Archea?
Star shaped– genus Stella
Rectangular flat– genus Haloarcula
Triangular

70) EPS stands for?
a) Efficient polymeric substance
b) Extracellular polymeric substance
c) Extra polynumeric system
d) Epilepsy puss syndrome

71) Glycocalyx as_____ Protects cells within it. Facilitates communication amongst cells
a) Protecter
b) Biofilm
c) Facilitater
d) polymeric substance

72) _____Long filamentous structure that propels bacteria?
a) Chrolella
b) Algae

c) Flagellum
d) Basillus

73) How bacteria classified based on flagella?
Atrichous: No flagellum
Peritrichous: Distributed over the entire cell
Polar: Monotrichous

74) _____Used for attachment to surfaces or epithelial cells Singular____?
a) Coccus, cocci
b) Fimbriae, Fimbria
c) Bassilus, bassili
d) Flagellum, Flagela

From topics 17 to 20

75) Cell wall is the most important structure in prokaryotes that provides____?
a) Shape of bacteria
b) Protection from osmotic lysis
c) Pathogenicity
d) All of these

76) In ____ A Danish microbiologist _____ explain the Gram staining with difference of Gram –ive and +ive?
a) 1988, Graham
b) 1996, Joddy Hopps
c) 1898, Chief Bogo
d) 1899, Nickoloudus john

77) TEM means?
Transmission electron microscopy

78) TEM revealed true difference of Peptidoglycan that its ____in G +ive cells &___ in G –ive cells
a) Thin, Thick
b) Thick, Thin
c) Motile, Non motile
d) Non Motile, Motile

79) Name the two suger molecules that Peptidoglycan Structure is made up of?
N-acetylglucosamine & N-acetylmuramic acid

80) The Two suger molecules of Peptidoglycan Structure form the ____of the cell wall?

a) Shape
b) Thickness
c) Arrangement
d) Back bone

81) N-acetylglucosamine & N-acetylmuramic acid Molecules range from____molecules all together.
a) 5 – 75
b) 10 – 65
c) 20 – 80
d) 5 – 65

82) Penicillin interferes with _____bridges
a) Inter
b) Cross
c) Linked
d) All of these

83) LPS is abbreviated by ____?
a) Leuco primary saccharide
b) Lipopolysaccharide
c) Linked pencillin system
d) Leucopolysacchride

84) Which of the following is a component of LPS____?
a) Lipid A
b) Core polysaccharide
c) O side chain
d) All of these

85) In LPS the Lipid A is like an endotoxin which results in ____?
a) Fever
b) Vasodilation
c) Shock
d) All of these

86) What is function of Core polysaccharide & O side chain in LPS?
Core polysaccharide gives Structural support & O side chain is just like an Antigenic like teichoic acid in gram positive bacteria

87) In gram sataining gram +ive bacteria gives ___color & -ive gives ____color
a) Red, violet
b) Pink, blue
c) Red, pink
d) Both a and b

88) Describe Comparison of G-ive and G+ive Bacteria?

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1	Characteristics	Gram Positive	Gram Negative
2	Gram reaction	Retain CV dye	Decolorized
3	Peptidoglycan layer	Thick	Thin
4	Teichoic Acid	Present	Absent
5	Periplasmic Space	Absent (generally)	Present
6	Outer membrane	Absent	Present
7	LPS	Virtually none	High
8	Lipid and lipoprotein contents	Low	High
9	Flagellar structure	2 rings	4 rings
10	Toxin Produced	Exotoxins	Exo and endotoxins
11	Susceptibility to penicillin	High	Low
12	Overall resistance	High	Low

89) Gram positive cell wall is ____as compared to the gram negative cell wall?
a) Thick
b) Thin
c) Motile
d) Non motile

90) Lysozyme ____the sugar-derived backbone?
a) Sustain
b) Breaks
c) Grow up
d) Cultivate

91) Some bacteria lack cell wall are called as____?
a) Lysozyme
b) Protoplast
c) L-form bacteria
d) All of these

92) Which of the following is a function of cell membrane?
a) Photosynthesis

b) Lipid synthesis
c) Cell wall devision
d) All of these

93) Explain the Types of Membrane Proteins?

Peripheral Proteins are Loosely connected to the membrane. They are Soluble in water

Integral proteins are Imbedded in the membrane. They are also Insoluble in water & Amphipathic in nature

Some act as receptors

94) which of the following is the best example of Photosynthesis by Cell Membrane?

a) Chromatophores
b) Thalakoids
c) Both a or b
d) None of these

From topics 21 to 26

95) By Moving with the concentration gradient the Substances move from area of _____concentration?

a) Low concentration to high
b) High concentration to low
c) High concentration to very high
d) Low concentration to very low

96) _____are the Movement against the concentration gradient From low concentration to high concentration & Requires energy (ATP)

a) Active Movement
b) Passive Movement
c) Specific Movement
d) Nonspecific Movement

97) In Facilitated diffusion Integral proteins as channels or carriers are Called____?

a) Transporters
b) Permeasess
c) Polemerases
d) Both a or b options

98) Nonspecific transporters are Ions while specific transporters Change in shape for example_____?

a) Carbohydrates
b) Vitamins
c) Lipids
d) Proteins

99) In Facilitated diffusion ___molecules degraded by extracellular enzymes _____molecules then bind transporters
a) Small, Larger
b) Large, Smaller

100) Water molecules can pass through lipid bilayer by simple diffusion or though_____?
a) Aquaporins
b) Artificial pipes
c) Channels
d) All of these

101) Define Osmotic Pressure?
A Pressure required to prevent the movement of water into a solution containing some solute.

102) Difference b/w Passive diffusion & active diffusion?
Passive: No ATP used, Conc. Gradient dependent llike Simple diffusion, Facilitated diffusion
Active diffusion: ATP used, Go against concentration gradient like Active transport, Group translocation

103) In active movement requiring ATP Which Nutrients move from lower conc to higher conc?
Na+,K+, H+, Ca2+, and Cl–, Amino acids, Simple sugars

104) How much water used in prokaryotic cytoplasm?
a) 30%
b) 50%
c) 80%
d) 96%

105) Gas Vacuoles are Aquatic bacteria that Maintain buoyancy for obtaining_____?
a) Nutrients
b) O2
c) Light
d) All of these

106) It Surrounded by invaginations of plasma membrane and Present in G negative bacteria Act like magnet Decompose H2O2
a) Gas vacuoles
b) Magnetosomes
c) Metachromatic Granules
d) Volutin

107) What are endospores?
Endospores are Specialized resting cells Highly durable, dehydrated forms of bacteria. They Formed inside a bacterium Survive extremes of temperature. they are Lack of water, Radiation& Toxic environment

108) Spore is a specialized cell Forms when nutrition gets ___?
a) Duplicated
b) Deleted
c) Depleted
d) All of these

109) Give some Examples of coenzymes?
– NAD+, – NADP+, – FMN and FAD

110) In Oxidative Phosphorylation the Electrons transferred to ____?
a) NAD+
b) FAD
c) NADP+
d) Both a and b

111) What is Glycolysis?
It is the oxidation of glucose to pyruvic acid with the production of some ATP and energycontaining NADH.

112) Define kreb cycle?
It is Oxidation of acetyl CoA to carbon dioxide, with the production of some ATP and energy-containing NADH and FADH2.

From topics 26 to 30

113) _____Uses light as a source of energy.
a) Chemoautotrophs
b) Photoautotrophs
c) Both of these
d) None of these

114) Green bacteria are____?
a) Oxygenic
b) Non-oxygenic

c) Both of these
d) None of these

115) Lipids and proteins can enter at various points in the____?
a) Glycolysis
b) Krebs cycle
c) Fermentation
d) Both a and b options

116) Microbes classified on the basis of _____Requirements?
a) Bacterial
b) Physical
c) Nutritional
d) Chemical

117) What are Physical requirmments and Chemical Requirements for bacterial growth ?
physical: Temperature, pH & Osmotic Pressure
Chemical: Source of C, N, S, P, O,Trace elements Growth factors

118) How bacteria Classified based on temperature?
Psychrophiles: – Cold loving: 15 0C
Mesophiles: 25 – 40 0C – Moderate temp loving Optimum: 37 0C
Thermophiles: 50 – 60 0C – Heat loving– Important in organic compost piles

119) Psychrotrophs Grows best at _____ Can grow at refrigeration temperature?
a) 18 – 30 0C
b) 20 – 30 0C
c) 30 – 40 0C
d) 30 – 100 0C

120) Food spoilage occurs between
a) 5 - 60 0C
b) 18 – 30 0C
c) 30 – 40 0C
d) 10 – 100 0C

121) Most bacteria grow at _____ pH.
a) 5.5 to 6.5
b) 6.5 to 7.5
c) 7.5 to 8.5
d) 6.7 to 7.0

122) Osmotic pressure is necessary for integrity of ____?
a) Microbes
b) Viruses

c) Bacteria
d) Fungus

123) Chemoheterotrophs Use C from energy sources like____?
a) Proteins
b) Carbs
c) Lipids
d) All of these

124) Sulfur and phosphorus needed for _____?
a) Nucleic acids
b) Sulphuric acids
c) Phosphoric acids
d) All of these

125) _____during Kreb's cycle oxidized via ETC and generate 36 or 38 ATP molecules
a) NADH
b) FADH2
c) NAD
d) Both a and b

126) Anaerobes do not have SOD, so cannot grow in the presence of ___?
a) CO2
b) O2
c) CO3
d) None of these

From topic 31 to 36

127) _____is a nutrient material for growth Some of them grow easily, other do not grow at all?
a) Culture medium
b) Growth medium
c) Nutritional medium
d) Bacterial medium

128) For growth initiation which of the following Microbe introduced into a culture medium?
a) O2
b) Complex carb
c) Petri plates
d) Inoculum

129) Pure culture is easy to obtain by _____ the organisms

a) Displaying
b) splashing
c) banding
d) streaking

130) who was the first to develop pure culturing technique.
a) Robert Koch
b) Robert hook
c) H Metheleda
d) Emiet Autiton

131) In how many categories The media that we used for bacterial or microbial growth is classified? Also mention their names?
Chemical composition, physical nature, function

132) Nutrient agar+ blood=?
a) Nutrient blood
b) Blood agar
c) Nutrient agar blood
d) None of these

133) β-hemolysis includes _____?
a) clear zone
b) complete hymolysis
c) greenish halo
d) both a and b

134) α-hemolysis contained ____?
a) Greenish halo
b) Partial hemolysis
c) Clear zone
d) Both a and b

135) In Classification of Media Chemical composition contained defined and complex media what is its mean?
Defined or synthetic means we know the organism
Complex mean an organism is not known or is fastidious

136) which of the following things are required for the production of MacConkey Agar?
a) Nutrient agar
b) Bile salt
c) CV
d) All of these

137) _____is A progeny of single cell?
a) Pure Culture

b) Culture medium
c) Inoculum
d) Nutrient blood

138) In pour plate method how many colonies are used?
a) 10 to 100
b) 20 to 200
c) 30 to 300
d) 40 to 400

139) What mehods are included in pure culture? Mention their names
Streak Plate method, Spread plate and pour plate method

140) Which of the following option refers to the batch culture of bacterial growth?
a) Nutrients depleted
b) Waste products increased
c) Waste products decreased
d) Both a and b

141) How the Generation time Can be calculated from the exponential phase of growth?

$$k = \frac{\log 10^9 - \log 10^3}{(0.301)(10\ \text{hr})} = \frac{9-3}{3.01\ \text{hr}} = 2.0\ \text{generations/hr}$$

$$g = \frac{1}{2.0\ \text{gen. hr}} = 0.5\ \text{hr/gen. or } 30\ \text{min/gen.}$$

142) How many phases are there in growth curve of bacterial growth?
a) 2
b) 3
c) 4
d) 5

143) Tell the names of phases in the growth curve of bacteria; growth?
Lag phase: initiatial
Log phase: first increasing phase
Stationary phase: constant phase (in my words)
Decline phase: decreasing phase

144) Which of the following method is a Viable counting method?
a) Spread plate
b) Pour plate

c) Pure culture
d) Both a and b

145) Direct methods are used for _____?
a) Microbial count
b) Dry mass
c) Most probable number
d) All of these

146) Indirect methods include?
a) Dry mass
b) Spectrophotometer
c) Most probable number
d) All of these

147) _____is Removing of all microbial life including spores?
a) Sterilization
b) Disinfection
c) Antisepsis
d) Degerming

148) _____is Removing of pathogens significantly?
a) Sterilization
b) Disinfection
c) Antisepsis
d) Degerming

149) _____is Removing of pathogens from living tissue?
a) Sterilization
b) Disinfection
c) Antisepsis
d) Sanitization

150) _____is Removing of microbes from a limited area (injection site)?
a) Degerming
b) Disinfection
c) Antisepsis
d) Sanitization

151) _____is Lowering of microbial counts on eating utensils?
a) Degerming
b) Disinfection
c) Antisepsis
d) Sanitization

152) Difference b/w Sepsis and Asepsis?

Sepsis is a Microbial contamination in living organisms and Asepsis is the Absence of significant contamination like we use Aseptic surgery techniques to prevent microbial contamination of wounds

From topics 36 to 40

153) Microbial death rate observe that _____?

a) Organisms don't die at once
b) Death rate is constant
c) Birth rate is constant
d) Both a and b

154) Decimal reduction time is the Time required to kill _____at a given Temperature?

a) 90% of microbes
b) 80% of bacteria
c) 70% of microbes
d) 60% of bacteria

155) Decimal reduction time is also Called ____which is time to drop the value by 1 log or 10-fold

a) D value
b) E value
c) G value
d) Z value

156) What are the Factors that influence microbial death rate

No. of microbes, Environment, Organic matter, pH Temperature, Exposure time, Microbial characteristics, Concentration of antimicrobial.

157) Acidic pH _____ the death rate?

a) Decreases
b) Increases
c) Has no effect on
d) constant

158) What is meant by TDP & TDT?

Thermal death point & Thermal death time

159) Thermal death point is the Lowest temperature at which all cells in a culture are killed in ____?

a) 10 minutes
b) 30 minutes
c) 45 minutes
d) 55 minutes

160) _____is the minimum Time during which all cells in a culture are killed at given temperature.

a) TDP
b) TDT
c) TPT
d) DTP

161) In microbial control the Physical method includes_____?

a) Heat
b) Radiation
c) Flaming
d) Both a and b

162) How a dry Heat is classifies?

There are two types of dry heat which are incineration in which objects burns and dry oven in which we just provide heat but not burn the objects

163) For achieving sterilization if we use hot air the temperature has to achieved_____& time must be _____?

a) 100 C, 1hr
b) 90 C, 2 hr
c) 150 C, 1 hr
d) 170 C, 2 hr

164) Pasteurization is a way to_____heat?

a) Dry
b) Moist
c) Change
d) Increase

165) What should be the initial temperature of pasteurization?

a) 63°C for 30 min
b) 53°C for 10 min
c) 36°C for 35 min
d) 60°C for 40 min

166) What should be the maximum temperature of pasteurization?

a) 72°C for 15 sec
b) 53°C for 10 min
c) 60°C for 40 sec
d) 63°C for 30 sec